

CANOAS, Brazil

WMO: 839670

Lat: 29.946S

Lon: 51.144W

Elev: 26

StdP: 14.68

Time Zone: -3.00 (SAB)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 37.7	(c) 40.8	(d) 33.6	(e) 28.2	(f) 44.0	(g) 35.9	(h) 30.9	(i) 45.6	(j) 22.2	(k) 56.9	(l) 19.8	(m) 56.4	(n) 1.7	(o) 330	(p) 0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.8	96.5	76.0	93.2	75.5	90.4	74.7	79.6	89.4	78.3	87.3	77.1	85.4	8.8	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.1	141.3	83.0	75.6	134.0	81.4	75.0	131.4	80.8	43.1	89.4	41.7	89.1	40.4	84.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	18.6	16.6	32.2	101.3	2.0	4.0	30.8	104.2	29.6	106.5	28.4	108.7	27.0	111.6
			31.8	82.4	3.2	1.9	29.5	83.8	27.6	84.8	25.9	85.9	23.6	87.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.2	78.1	78.5	74.0	69.2	63.2	58.1	57.0	61.0	64.5	68.0	71.4	75.8
	DBStd	9.18	4.62	4.75	4.64	5.41	5.45	6.75	7.13	7.42	5.94	5.24	5.00	5.06
	HDD50	31	0	0	0	0	0	9	18	4	0	0	0	0
	HDD65	895	0	0	2	22	100	224	262	170	79	28	7	1
	CDD50	6665	873	797	743	576	409	253	234	345	437	557	642	799
	CDD65	2051	408	377	280	147	44	18	13	46	65	120	199	334
	CDH74	16891	3654	3351	2055	979	264	145	186	526	501	793	1570	2867
	CDH80	6476	1573	1460	698	292	40	26	35	195	181	246	541	1189
Wind	WSAvg	7.0	8.1	7.3	7.1	5.8	5.3	5.0	5.6	6.4	7.9	8.3	8.7	8.6
Precipitation	PrecAvg	54.7	4.4	4.4	4.3	4.0	4.2	5.2	5.4	5.2	5.7	5.1	4.0	4.3
	PrecMax	72.4	9.4	9.9	11.9	11.6	17.3	14.4	12.2	13.0	11.7	12.1	11.3	8.8
	PrecMin	28.0	0.4	0.9	0.6	0.0	0.1	0.3	0.7	0.5	0.6	0.8	0.3	1.2
	PrecStd	9.3	2.1	1.9	2.2	2.5	3.0	3.0	2.7	2.8	2.6	3.0	2.4	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.5	102.1	95.1	91.2	84.3	84.0	84.2	91.0	92.8	94.6	93.6	98.7
		MCWB	79.4	79.0	75.6	74.4	71.8	69.5	66.4	68.5	71.8	74.3	73.7	77.2
	2%	DB	95.1	95.4	90.5	87.5	80.4	78.7	79.6	85.9	84.5	86.0	89.8	93.5
		MCWB	76.7	78.4	74.7	72.9	67.8	67.6	65.4	67.9	70.1	71.6	72.9	74.8
	5%	DB	91.5	91.7	87.5	83.8	76.9	73.3	73.7	80.2	78.8	82.0	86.2	89.9
		MCWB	76.0	77.0	74.0	71.1	67.3	65.3	63.4	67.1	67.9	69.8	71.2	74.3
	10%	DB	88.1	88.1	84.2	80.2	73.2	68.4	68.2	73.7	74.9	78.5	82.6	87.2
		MCWB	75.7	75.7	72.5	69.5	65.5	63.2	61.5	64.4	65.6	68.2	69.7	73.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.9	82.2	78.7	76.2	73.8	71.3	69.4	71.1	73.9	75.5	78.5	79.9
		MCDB	91.9	94.2	90.2	86.5	79.7	80.5	78.6	84.1	84.5	87.9	86.6	89.7
	2%	WB	79.4	79.8	76.8	74.3	70.2	68.8	66.7	69.1	71.0	72.8	75.9	78.1
		MCDB	89.2	89.2	86.3	82.9	75.8	76.0	75.3	80.9	81.3	82.4	84.1	87.4
	5%	WB	78.1	78.3	75.5	72.6	68.7	66.6	64.9	67.4	69.0	71.2	73.7	76.5
		MCDB	86.9	86.5	83.5	79.9	73.7	72.3	72.0	77.7	77.0	79.2	80.8	85.0
	10%	WB	76.8	76.9	74.2	71.2	67.0	64.4	62.8	65.5	67.1	69.5	72.2	75.2
		MCDB	84.4	84.3	81.0	76.9	71.7	68.5	67.5	73.3	72.9	75.7	79.0	82.9
Mean Daily Temperature Range	5% DB	MDBR	19.1	18.8	18.3	19.3	18.0	17.5	19.6	20.9	18.3	18.4	20.6	20.2
		MCDBR	24.3	23.8	23.6	24.1	24.0	25.0	28.4	31.1	26.8	25.3	26.7	25.8
		MCWBR	9.0	9.1	10.1	10.8	12.7	14.5	15.7	14.8	13.2	11.8	11.5	10.1
	5% WB	MCDBR	19.2	20.2	19.0	19.8	18.0	21.2	23.8	27.5	23.7	21.1	20.5	20.6
		MCWBR	8.4	8.5	8.8	9.8	10.3	12.6	14.3	14.0	12.3	10.7	9.8	8.8
Clear-Sky Solar Irradiance	taub		0.408	0.411	0.400	0.380	0.367	0.382	0.388	0.475	0.522	0.454	0.414	0.412
	taud		2.416	2.409	2.431	2.447	2.460	2.418	2.364	2.063	1.948	2.203	2.338	2.373
	Ebn at Noon		296	289	282	271	259	245	248	235	241	274	292	296
	Edh at Noon		40	39	36	32	28	28	31	46	57	47	43	42
All-Sky Solar Radiation	RadAvg		2023	1854	1568	1239	915	772	849	1052	1265	1572	1963	2065
	RadStd		142	121	106	100	87	57	97	90	125	173	206	167

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page